

Date	16 September 2026
Team ID	SWTID-2026-9627
Project Name	AI Fit Track API
Maximum Marks	2 Marks

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., User, Google Gemini AI API).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., 1.0 Authenticate User).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., D1 Users DB, D2 Workouts DB).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

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sequenceDiagram
    participant User as External Entity: User (Client / Postman / Thunder Client)
    participant Logic as 1.0 Authenticate User  
validates credentials, hashes password (bcrypt), issues JWT
    participant Logic2 as 2.0 Manage Workouts  
create/read/update/delete/search workout records for the authenticated user
    participant DB1 as D1 — Users DB (MongoDB)
    participant DB2 as D2 — Workouts DB (MongoDB)
    participant Logic3 as 3.0 Generate Workout Recommendation  
geminiService.js sends age/fitness goal/experience level to Gemini
    participant Logic4 as 4.0 Generate Fitness Insights  
geminiService.js sends aggregated workout stats to Gemini
    participant Gemini as External Entity: Google Gemini AI API
    participant User2 as External Entity: User (Client) receives final response

    User->>Logic: ↓ registration / login / workout / AI requests | ↑ JWT, JSON responses
    Logic->>DB1: ↓ ↑
    Logic2->>DB2: ↓ ↑
    Logic3->>Gemini: ↔ age, goal, experience | workout plan text
    Logic4->>Gemini: ↔ total workouts, avg duration, calories | insight text
    Gemini->>User2: ↓ structured JSON (recommendation / insight) returned to User
    User2->>User: External Entity: User (Client) receives final response
```

External Entity: **User (Client / Postman / Thunder Client)**

↓ registration / login / workout / AI requests | ↑ JWT, JSON responses

1.0 Authenticate User
validates credentials, hashes password (bcrypt), issues JWT

2.0 Manage Workouts
create/read/update/delete/search workout records for the authenticated user

↓ ↑

D1 — Users DB (MongoDB)

D2 — Workouts DB (MongoDB)

3.0 Generate Workout Recommendation
geminiService.js sends age/fitness goal/experience level to Gemini

4.0 Generate Fitness Insights
geminiService.js sends aggregated workout stats to Gemini

↔ age, goal, experience | workout plan text

↔ total workouts, avg duration, calories | insight text

External Entity: **Google Gemini AI API**

External Entity: **Google Gemini AI API**

↓ structured JSON (recommendation / insight) returned to User

External Entity: **User (Client)** receives final response

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